

HOMEWORK 10
ELEMENTARY LINEAR ALGEBRA
(MATH 2250-03), SPRING, 2019

CHECK: 05/03/2019

SECTION 6.1 - INNER PRODUCT, LENGTH, AND ORTHOGONALITY

1. Let $u = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$ and $v = \begin{bmatrix} 4 \\ 6 \end{bmatrix}$. Compute the following quantities:

- (a) $\|u\| - \|v\|$ (c) $u \cdot u$ (e) $\left(\frac{u \cdot v}{v \cdot v}\right)v$
(b) $\|u - v\|$ (d) $v \cdot u$ (f) $\left(\frac{1}{u \cdot u}\right)u$

2. Find the distance between $x = \begin{bmatrix} 10 \\ -3 \end{bmatrix}$ and $y = \begin{bmatrix} -1 \\ -5 \end{bmatrix}$.

3. Let W be a subspace of $\mathbb{R}^n$. Show that if $x \in \mathbb{R}^n$ is a vector which satisfies both $x \in W$ and $x \in W^\perp$, then $x = 0$.

SECTION 6.2 - ORTHOGONAL SETS

4. Determine whether

$$\left\{ \begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \end{bmatrix}, \begin{bmatrix} -1/2 \\ 0 \\ 1/2 \end{bmatrix} \right\}$$

is an orthogonal set. If it is, normalize the vectors in this set to produce an orthonormal set.

5. Compute the orthogonal projection of $\begin{bmatrix} 1 \\ 7 \end{bmatrix}$ onto the line through $\begin{bmatrix} -4 \\ 2 \end{bmatrix}$ and the origin.

6. Given a vector $u \neq 0$ in $\mathbb{R}^n$, let $L = \text{span}\{u\}$. Show that the mapping $x \mapsto \text{proj}_L(x)$ is a linear transformation.

SECTION 6.3 - ORTHOGONAL PROJECTIONS

7. Let $u_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ and $u_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$ and $y = \begin{bmatrix} -1 \\ 4 \\ 3 \end{bmatrix}$. Verify that $\{u_1, u_2\}$ is an orthogonal set. Then, find the orthogonal projection of y onto $\text{span}\{u_1, u_2\}$.

8. Let $v_1 = \begin{bmatrix} 2 \\ -1 \\ -3 \\ 1 \end{bmatrix}$, $v_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ -1 \end{bmatrix}$, and $z = \begin{bmatrix} 3 \\ -7 \\ 2 \\ 3 \end{bmatrix}$. Let $W = \text{span}\{v_1, v_2\}$ and find the best approximation to z by elements of W .

9. Let W be a subspace of $\mathbb{R}^n$ with an orthogonal basis $\{w_1, \dots, w_p\}$. Suppose $\{v_1, \dots, v_q\}$ is an orthogonal basis for $W^\perp$. Show that $\{w_1, \dots, w_p, v_1, \dots, v_q\}$ is an orthogonal basis for $\mathbb{R}^n$.